Oracle

Retrained on Homology-Restricted MegaScale

Zero-shot

Supervised, not trained on Megascale

Rectified gain @ k=96 (kcal/mol)

100
10
1
0

27.80 14.81 3.38
17.69 9.14 1.62
19.10 9.77 1.88
14.26 6.49 1.06
15.78 8.24 1.56
17.04 8.17 1.57
11.27 3.97 0.58
12.56 5.01 0.84
12.39 4.49 0.61
11.84 4.34 0.82
11.39 5.94 1.30

Predicted most stabilizing per domain

10
1
0

50.65 34.30 11.04
51.83 33.61 10.57
51.65 33.54 10.81
54.23 33.45 8.32
48.78 35.99 11.23
49.97 35.17 10.86
61.61 26.70 7.70
54.30 32.65 9.04
55.22 33.04 7.74
51.65 35.35 9.00
46.43 38.22 11.35

Top 96 per domain
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta

% conservation ≤ 25
25 < % conservation ≤ 75
% conservation > 75

Top 96 per Homology-Restricted MegaScale test domain (23 domains with the feature, 27,163 single mutants); gain = sum of max(ddG, 0) over every pick, whatever its predicted sign; boxes: interquartile range with darker colored median line; dots: domains; dark bars & annotations: mean over domains; dashed lines: oracle mean.